# Cameron Cox

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## EDUCATION

**The University of Texas at Austin**; Austin, Texas — August 2027

*Bachelor of Science in Computer Science + Minor in Robotics*

**Certifications:** *Electric Circuit Analysis UDEMY • Leadership Training CoachUT • Machine Shop Safety OH500 • Soldering, Laser Cutting, PCB Design/Printing, Woodshop Training, 3D Printing, and Manual Mill Machining at Texas InventionWorks*

## EXPERIENCE

**Robotics Software Engineering Intern**; Carrollton, Texas — May 2026 – August 2026

*REV Robotics*

- Designed a 154-test multithreaded Hardware-in-Loop (HIL) tester from the ground up for the A301 Motor, integrating 3 motors, a Systemcore, throughbore encoders, CAN Encoder adapters, a power supply, and a motioncore expansion board
- Employed agentic coding skills to implement a GitHub Action CI pipeline spanning 3 repos using a self-hosted Proxmox virtual machine to automatically compile, execute, and generate reports for the HIL tests and prevent regressions from being merged
- Reduced memory write times by 10 ms, reducing the wait from 15 to 5 ms (66.7%) and reducing test suite time by 1 minute
- Collected and analyzed data for current draw, motor performance, and thermal behavior using Python and MatplotLib
- Developed a Java production-testing application app for motoros and servo controllers testing factory products for function

**Undergraduate Course Assistant for CS 314: Data Structures & CS 303E: Elements of CS**; Austin, Texas — August 2025 – Present

*UT Austin Department of Computer Science*

- Lead 21 students through weekly coding reviews, office hours, and discussion sections covering data structures and algorithms
- Graded 500+ programming labs covering correctness, object-oriented programming, efficiency, and coding hygiene
- Collaborated with course faculty to clarify complex technical concepts and adapt instruction to student needs

**Robotics, Coding, and Game Dev Instructor**; Austin, Texas — May 2025 – August 2025

*Internal Drive Technology (iD Tech)*

- Instructed 60+ students in a multi-stage BattleBots® engineering design cycle covering gear ratios, C++, and mechanical design
- Engineered and facilitated BattleBots® challenges: integrating sensors, motor control, and drivetrain optimization
- Recognition by National Math Stars as top iD Tech Camp nationwide for instructional performance and communication

**Automation and Migration Engineer**; Kennedale, Texas — May 2024 – August 2024

*Behrens and Associates Technology, LLC*

- Implemented a deployment script for 30+ Docker containerized websites using Bash and containerized development
- Built a script to find compatible git commits through subprocess to prevent merge conflicts before committing to a branch
- Automated the processing of 300+ Azure Build Logs through a locally run ElasticStack using cURL commands and Azure DevOps

## LEADERSHIP & COMMUNITY INVOLVEMENT

**Vice President + Engineering Lead**; Austin, Texas

*Engineering and Computational Learning of Artificial Intelligence in Robotics (ECLAIR)*

- Actively managing 30+ active engineers across 7 robotics/AI projects; organizing project leads, staff, milestones, and resources
- Secured $15,000 in sponsorships, stipends, and grants from UTCS, TexelArts, Texas Robotics, Yaskawa, and ServiceNow
- Organized 25+ outreach/community events, including engineering expos, tech talks, robot hackathons, and work meetings

**Combat Robotics Team Lead**; Austin, Texas

*Experimental Division for Combat Robotics (EXD)*

- Designed, fabricated, assembled, and competed with DreamSiccle, a 1 lb. horizontal spinner, in 7 tournaments, winning 5 fights
- Built D.O.N.U.T. a 3 lb. translational-drift “meltybrain” combat robot using Fusion360, EasyEDA, and BambuStudio

## PROJECTS

C.R.A.C.K.L.E. – *First Place “Best Overall and Best Acronym Awards" out of 6 engineering projects*

- Built a 30-lb mobile manipulation robot, integrating a CIM-motor tank drivetrain, ESP32 control, robotic arm, custom gearbox, custom gripper, and an LLM-drive software architecture to achieve a self-growing codebase and autonomous navigation
- Developed an LLM-based self-growing code architecture that dynamically generates new robot capabilities, enabling the system to autonomously pick up objects, including a water bottle and phone.
- Built custom gearboxes via Onshape, 3D printing, and laser cutting to reduce drivetrain current 133 A to <25 A (81% reduction)
- Developed a custom C++ drivetrain API and integrated the robot's electronics through soldering and system-level wiring.

T.R.A.C.E. *First Place “Best Overall" award out of 6 engineering projects*

- Developed a closed-loop robotic platform using PID control and computer vision to balance and manipulate a ball
- Executed finite element analysis in Fusion360 to develop failure points, improve part geometry, and model the project

GPongT – *First Place “Best AI Hack” out of 5000 participants at HackTX ‘24*;

- Innovated an AI-based code generation architecture to analyze a codebase and generate supporting features at runtime
- Integrated the system into a game of pong to dynamically generate game levels with new challenges on subprocess thread

## OTHER SKILLS

CAD • Power Tools • PCB Design • Linux • A-64 Assembly • R • Bicep • CSS • KiCad • Trotec Ruby • CapCut • Microsoft Excel